Modern UX Sucks!

Reviving HCI Fundamentals for Web, Desktop, and AI





MOMENTUM
Developer Conference
powered by The Circuit

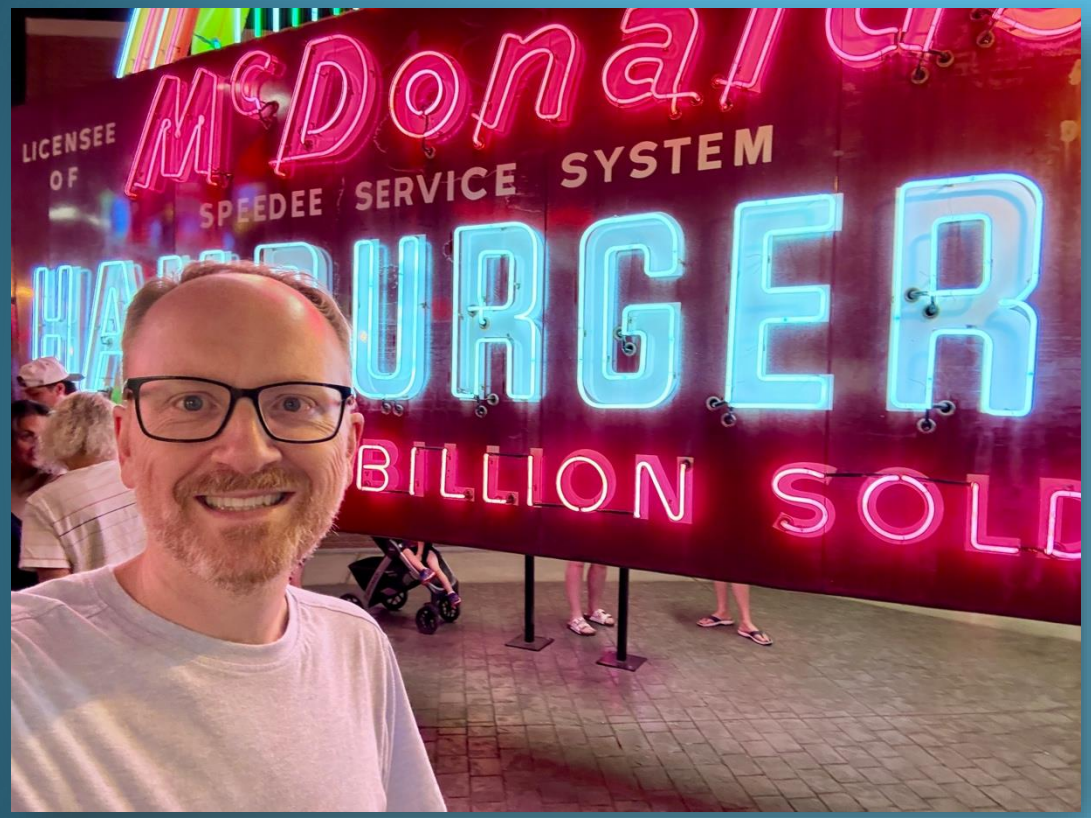
AI Disclosure

- Researched historical topics
- Generated a few images
- Proofreading
- Summarization of my notes

Modern UX SUCKS!

Reviving HCI Fundamentals for Web, Desktop, and AI


!!!!
BARISTA
COFFEES
Login
Login



Brandon Bruno

www.brandonbruno.com
/in/brandonbruno
Goshen, OH

- Software Developer
- Consultant
- Solutions Architect
- Sales Engineer
- Senior Solution Consultant



MOVIECLIPS.COM

Let’s Talk About **Charlotte…**



What is Human-Computer Interaction?

"Hey Siri, what is Human-Computer Interaction?"

Siri: "Alarm deleted."

"Explain Human-Computer Interaction to me."

Every LLM ever:

"Human-computer interaction is a sprawling, multidisciplinary field that sits at the chaotic intersection of computer science, behavioral psychology, cognitive science, industrial design, anthropology, and sociology. At its core, it is the study of how human beings interact with computational devices and the extent to which these technological systems are or are not designed for successful, intuitive, and efficient use

HCI is the study and understanding
of how **humans use computers**.

W I?



AN UPDATE UH-OH

Owners mourn spoiled food after firmware update bricks Samsung smart fridges

"So absurd and embarrassing …"

SCHARON HARDING – SEP 23, 2026 1:39 PM | 164


Steering Wheel
Seat
Arm Rest
Pedals





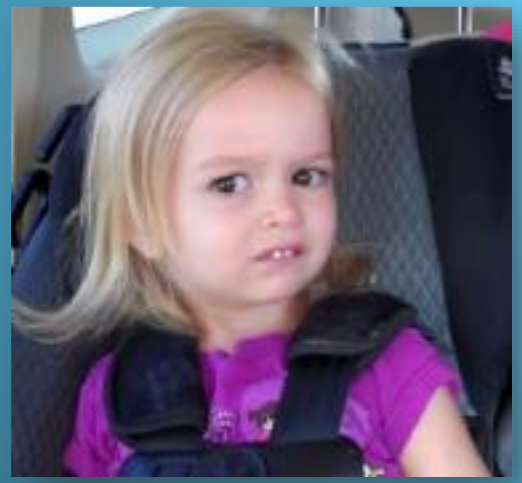




We Interact with our Senses

- **Touch**: mouse, keyboard, *touch*screens
- **Sight**: screens, printouts, lights, VR
- **Hearing**: sounds, beeps, music

- **Taste:**



- **Smell:**



Why does HCI matter today?





HCI Matters Because...



Universal Access

- Benefit humans
- Accessibility



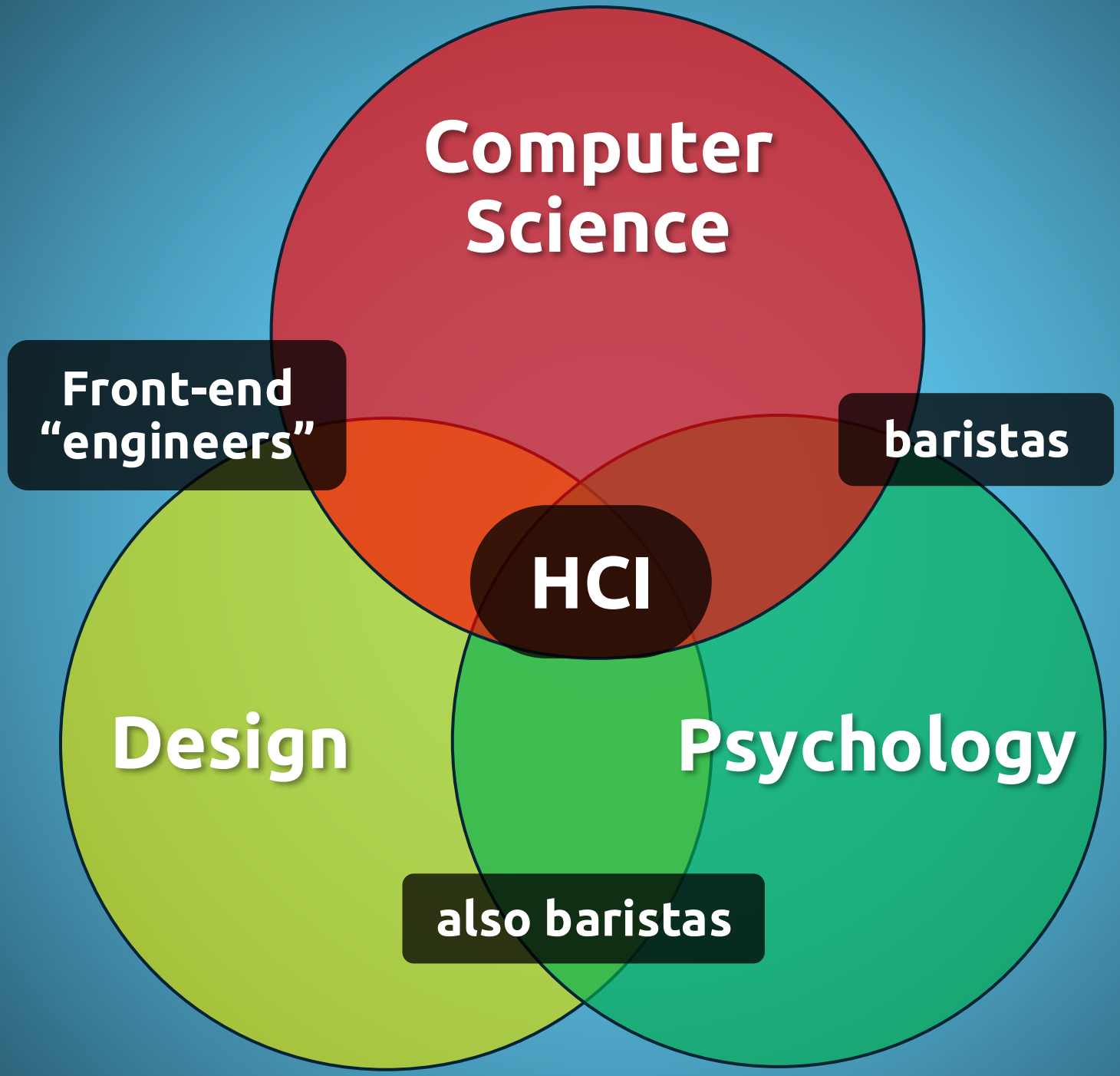
Increased Efficiency

- Move faster
- Reduce errors
  - (my $10k mistake)
- Safer outcomes



Business Value

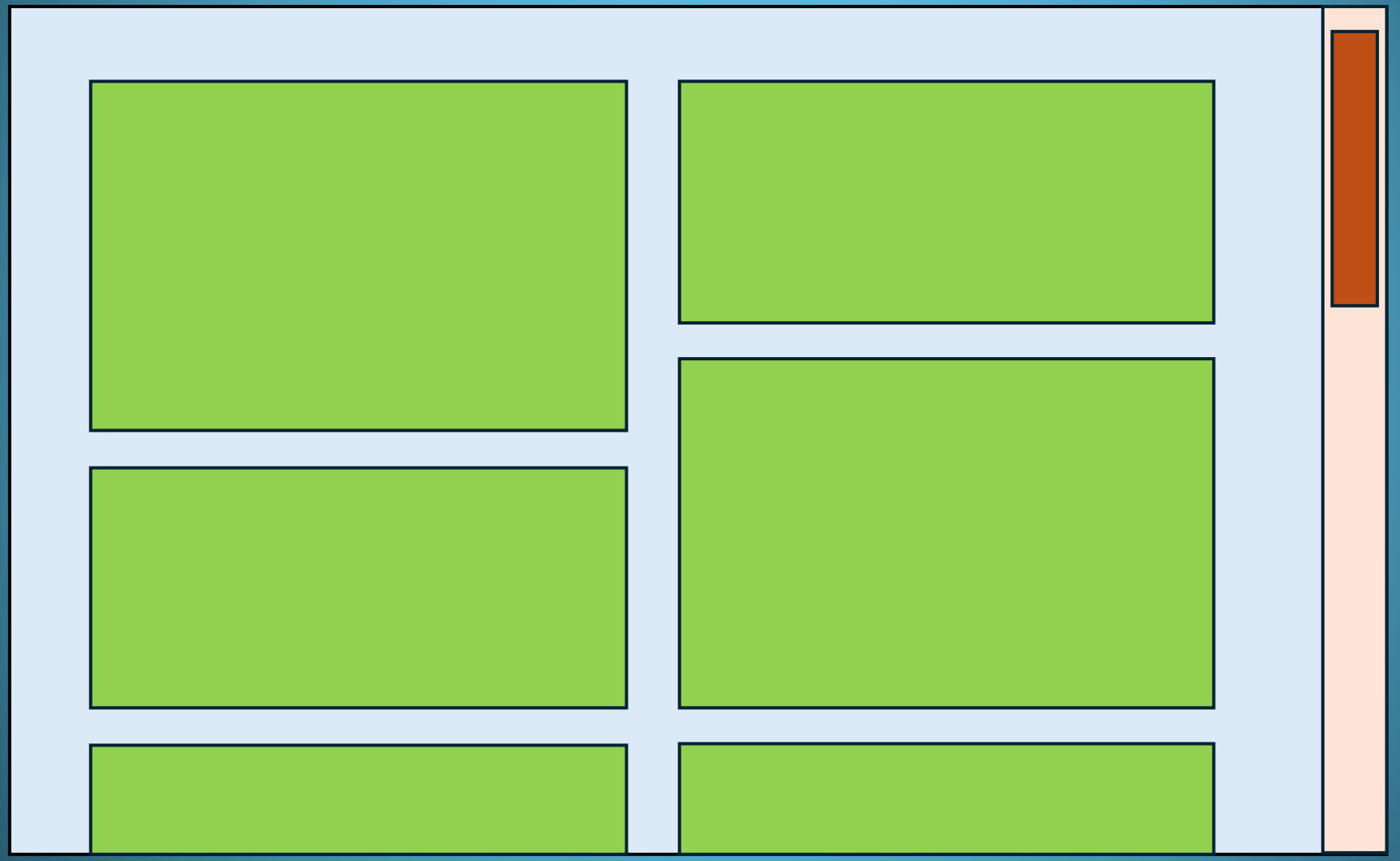
- User retention
- Satisfaction
- Reduced customer support


Computer Science
Front-end “engineers”
baristas
HCI
Design
Psychology
also baristas


Second Edition
Human-Computer Interaction
An Empirical Re
I. Scott
INTERACTION DESIGN
ROGERS • SHARP • PREECE
beyond humc
RESEARCH METHO
HUMAN-COMPU
INTERACTIO
Second Edition
Human Computer Interaction
User Behavior Analy
Interface Evaluation
and Usability-driven
design decisions
ALAN DIX, JAN
GREGORY D. ABOWD,
HUMAN–CO
INTERAC
THIRD EDITION
PEARSON
Prentice Hall
OXFORI
INTRODUCT
HUMAN-CO
INTERACTION
KASPER HORNBÆK
PER OLA KRISTENSSON | ANTTI OULA
SOLIS
HUMAN-C
INTERA
THE FUNDAMENTALS MADE EASY!
O'REILLY
Third Edition
Designing Interfaces
Patterns for Effective Interaction Design
Jenifer Tidwell,
Charles Brewer &
Aynne Valencia

Discoverability	**Is the UI obvious? Are actions easy to find?**
Feedback	**What was the result of an action? Success or failure?**
Consistency	**Re-use of patterns; user doesn't have to re-learn**
Affordance	**Design of elements imply their use and purpose**
Cognitive Load	**The processing & memory burden placed on a user**

Discoverability



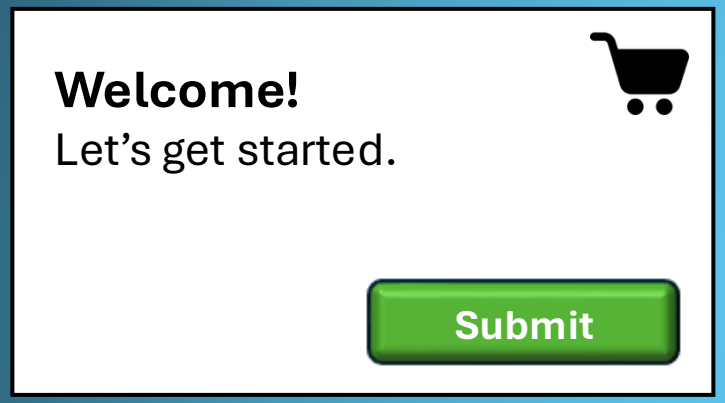
Feedback





Consistency

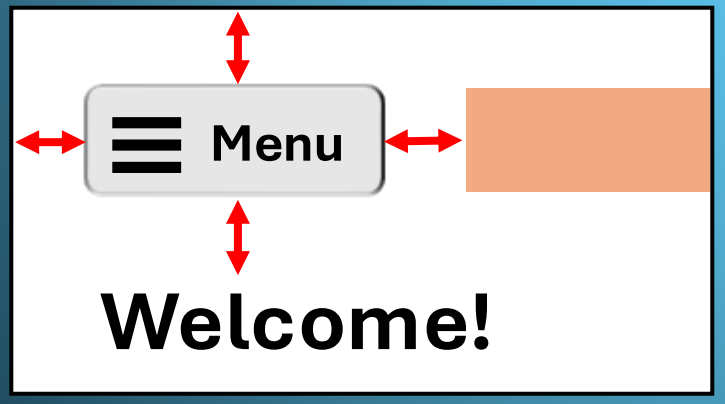
Internal



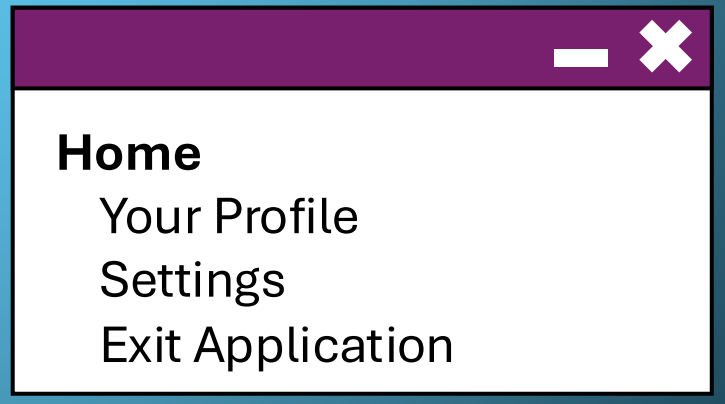
External



Visual



Functional



Affordance



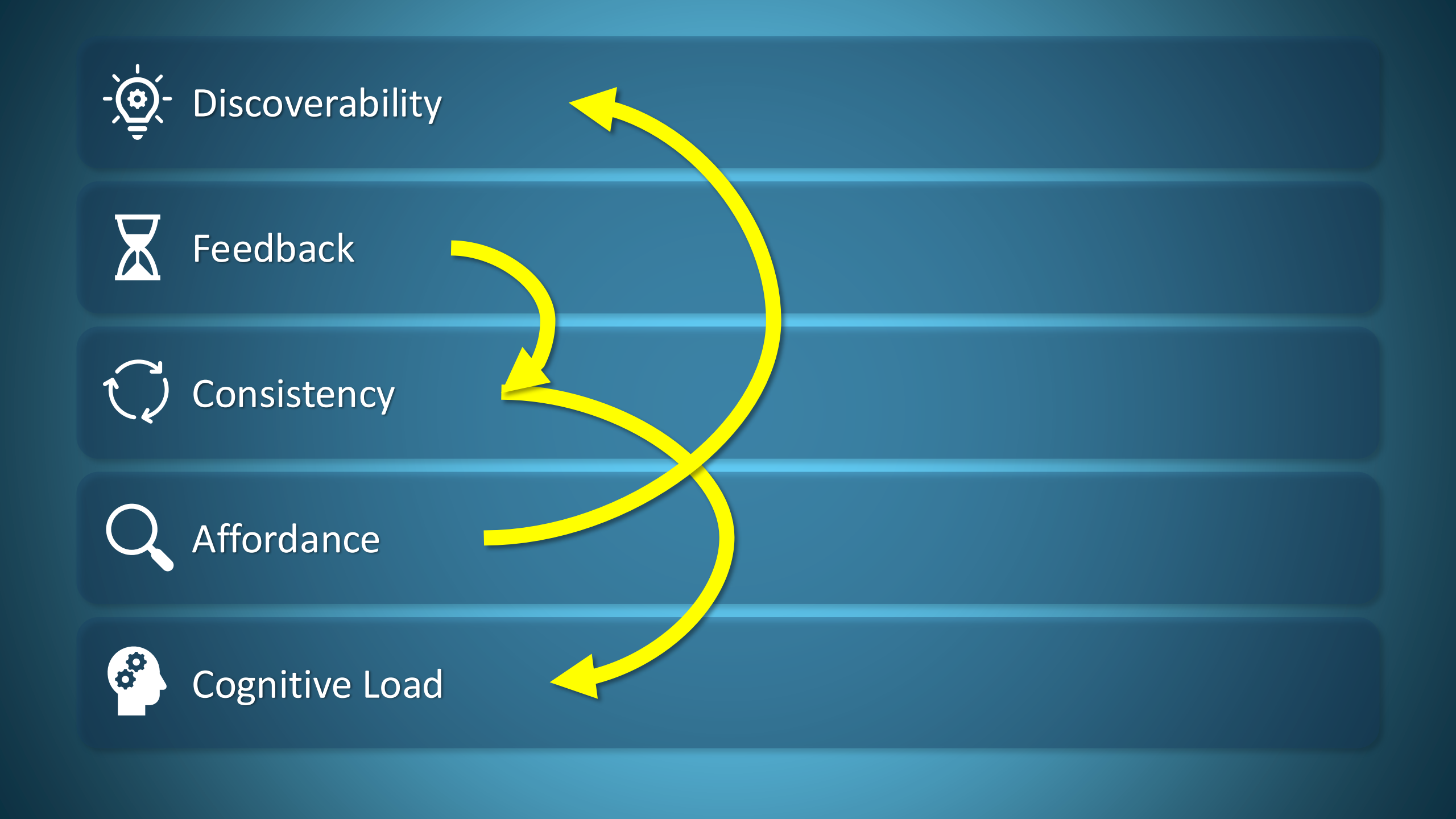


Brandon Bruno

Cognitive Load



Discoverability	**Is the UI obvious? Are actions easy to find?**
Feedback	**What was the result of an action? Success or failure?**
Consistency	**Re-use of patterns; user doesn't have to re-learn**
Affordance	**Design of elements imply their use and purpose**
Cognitive Load	**The processing & memory burden placed on a user**


Discoverability
Feedback
Consistency
Affordance
Cognitive Load

How do we interact with computers?



ArnoldReinhold, CC BY-SA 3.0, via Wikimedia Commons

Skeuomorphism



 = dispose

 = mail

 = save spot

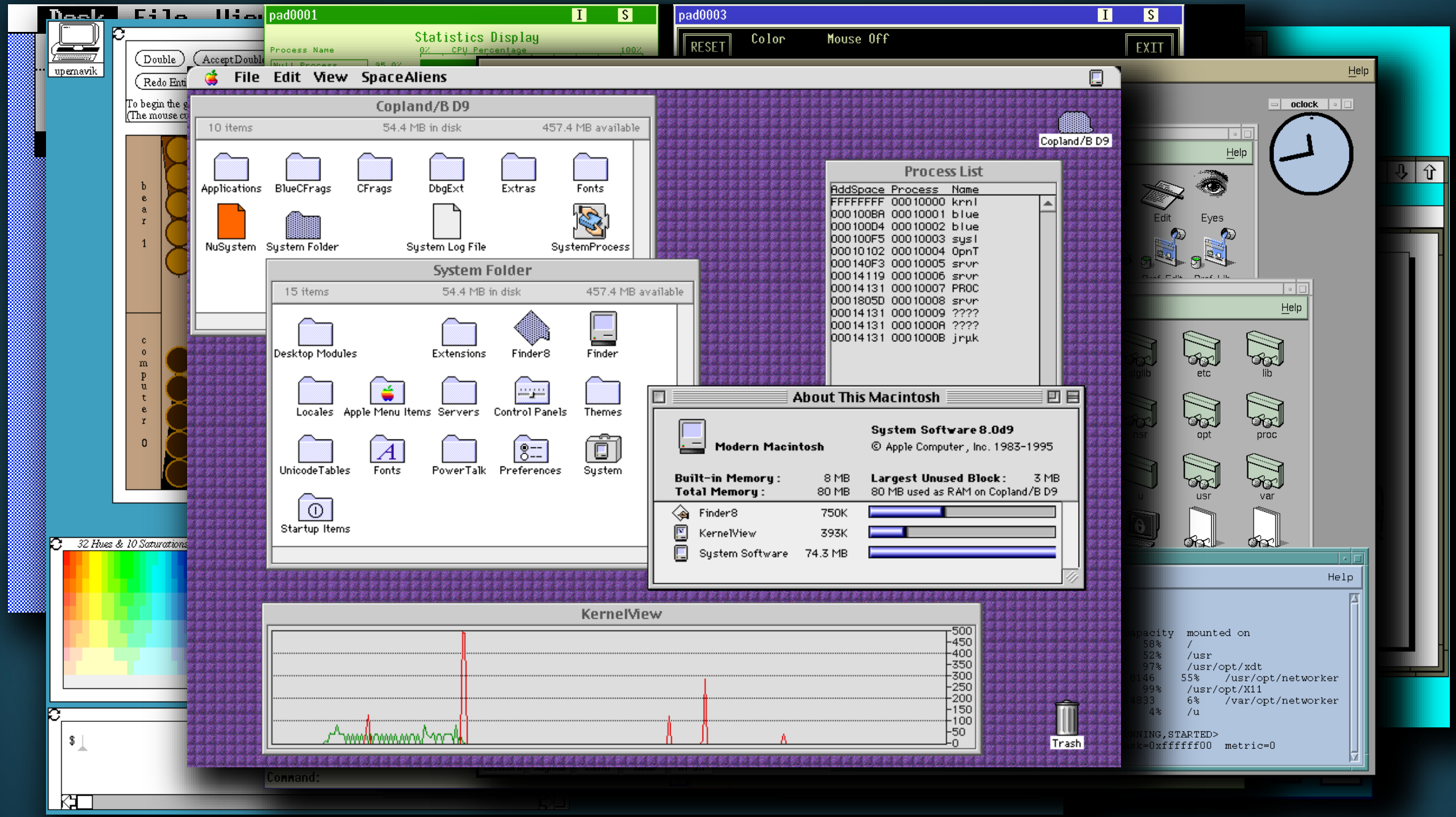
 = notes

 = save

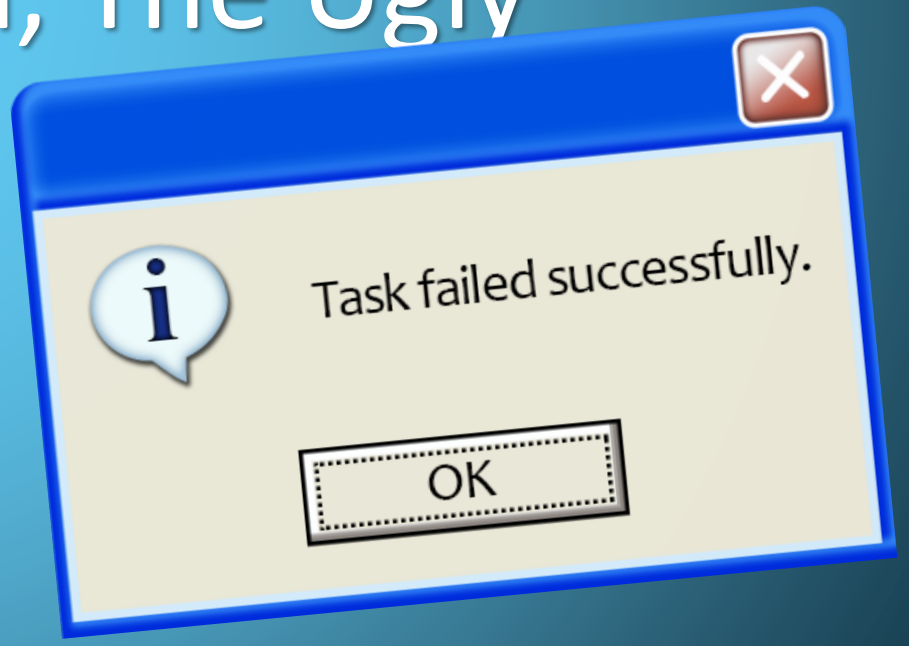
Skeuomorphism = Metaphors

 someone else's computer?

 a menu?


pad0001
Statistics Display
Process Name
CPU Percentage
pad0003
RESET
Color
Mouse Off
EXIT
File Edit View SpaceAliens
Copland/B D9
10 items
54.4 MB in disk
457.4 MB available
Applications
BlueCFrags
CFrags
DbgExt
Extras
Fonts
NuSystem
System Folder
System Log File
SystemProcess
System Folder
15 items
54.4 MB in disk
457.4 MB available
Desktop Modules
Extensions
Finder8
Finder
Locales
Apple Menu Items
Servers
Control Panels
Themes
UnicodeTables
Fonts
PowerTalk
Preferences
System
Startup Items
Process List
AddSpace Process Name
FFFFFFFF 00010000 krnl
000100BA 00010001 blue
000100D4 00010002 blue
000100F5 00010003 sysl
00010102 00010004 OpnT
000140F3 00010005 srvr
00014119 00010006 srvr
00014131 00010007 PROC
000180SD 00010008 srvr
00014131 00010009 ????
00014131 0001000A ????
00014131 0001000B jrµk
About This Macintosh
Modern Macintosh
System Software 8.0d9
© Apple Computer, Inc. 1983-1995
Built-in Memory : 8 MB
Total Memory : 80 MB
Largest Unused Block : 3 MB
80 MB used as RAM on Copland/B D9
Finder8 750K
KernelView 393K
System Software 74.3 MB
KernelView
500
450
400
350
300
250
200
150
100
50
0
Trash
Command:
Help
oclock
Edit
Eyes
etc
lib
opt
proc
usr
var
mounted on
/usr
/usr/opt/xdt
/usr/opt/networker
/usr/opt/X11
/var/opt/networker
/u
Double
Redo
upernavik
32 Hues & 10 Saturations

~~The~~ Good, the Bad, The Ugly





“A person is smart.

Designers

~~People~~ are dumb, panicky, dangerous animals…”

Justin Bieber
@justinbieber

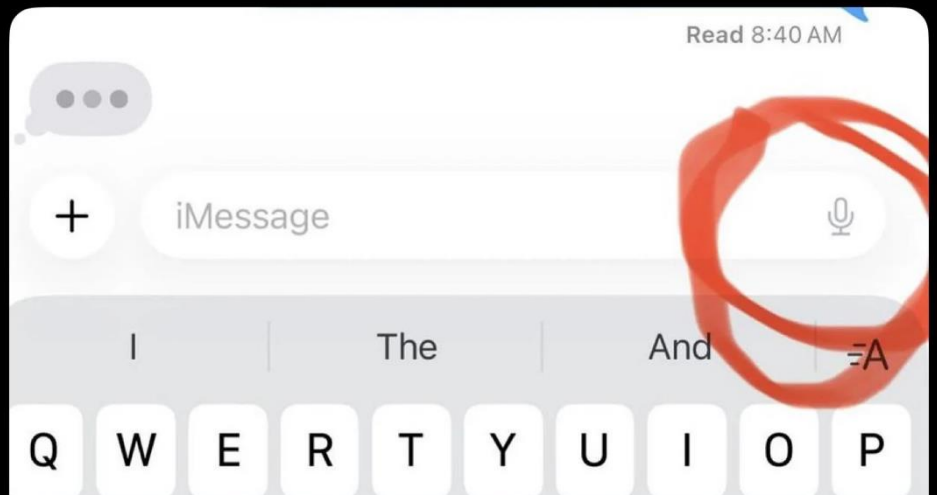
𝕏.com

If I hit this dictation button after sending a text and it beeps and stops my music one more time,

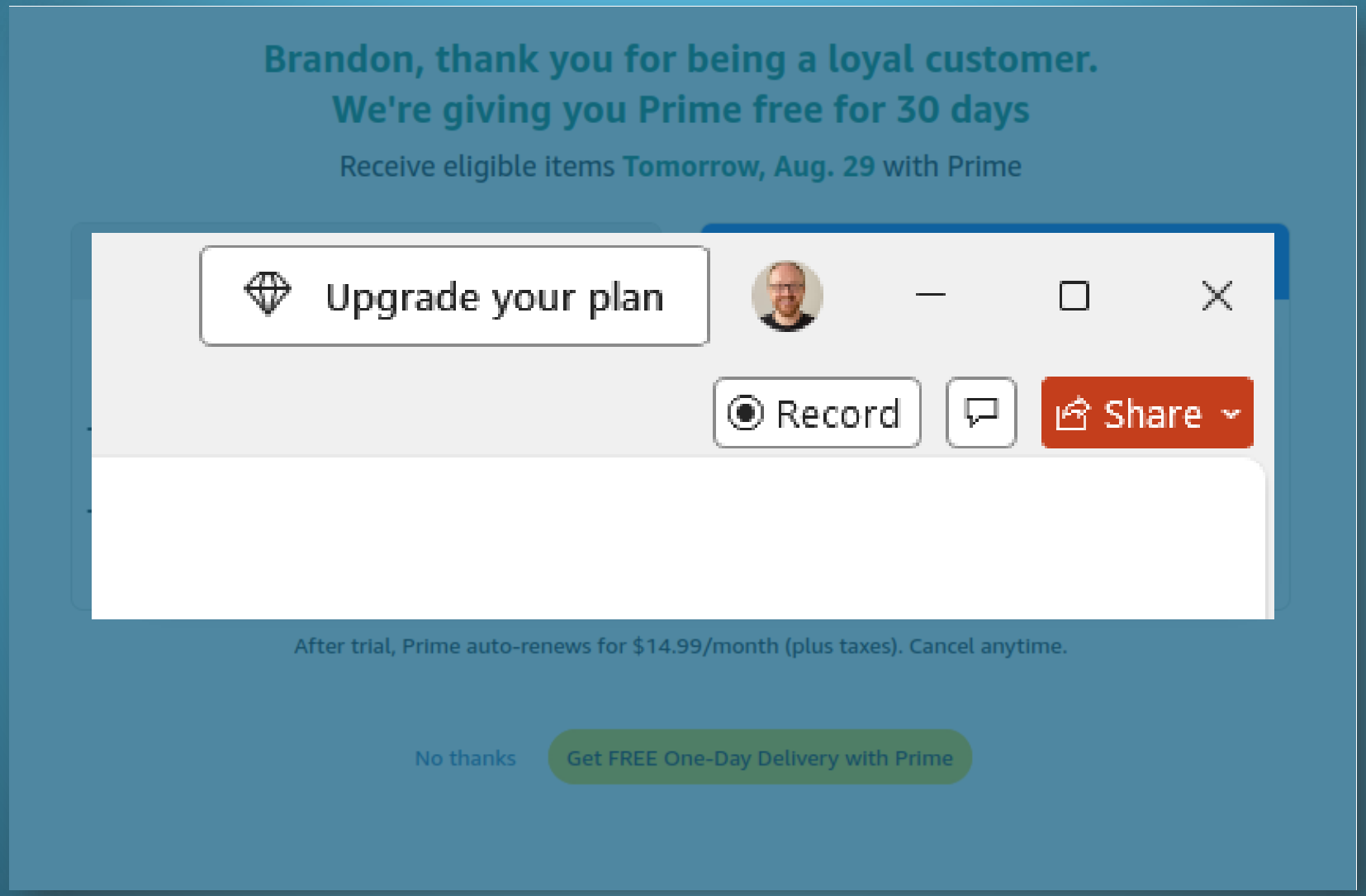
I’m gonna find everyone at apple and put them in a rear naked choke hold

Even if I turn off dictation I somehow hit the voice note thing

The send button should not have multiple functions in the same spot



Dark Patterns (all things $$$)



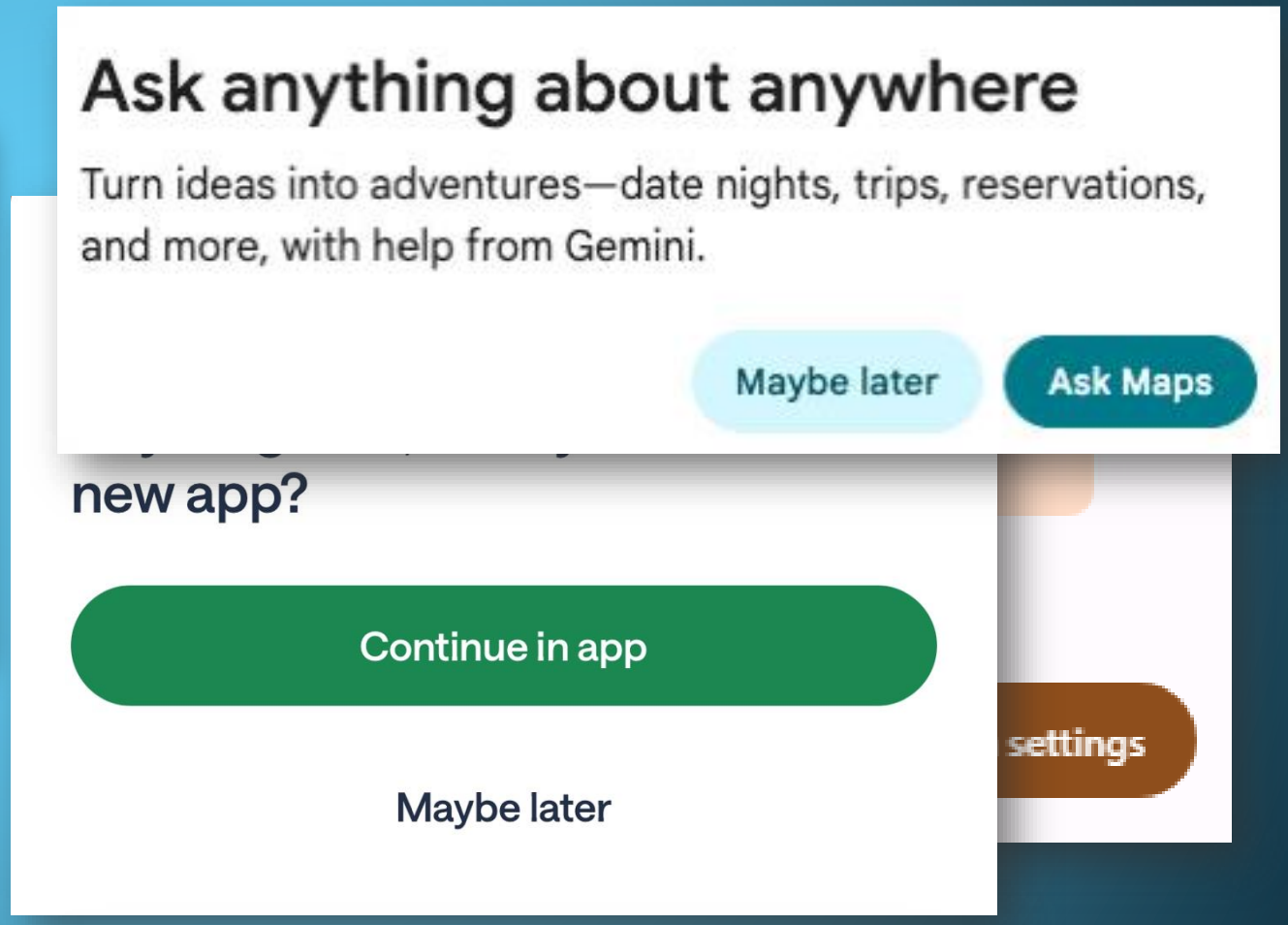
Scroll Hijacking



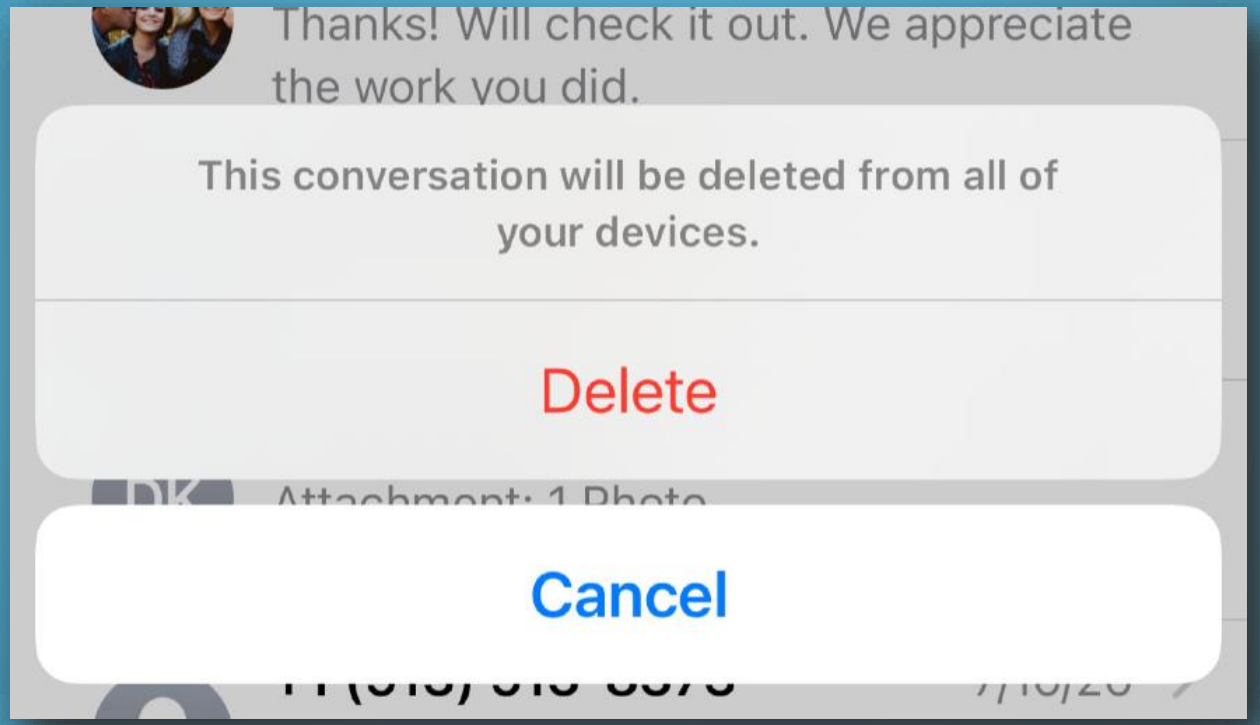
Consent Gone Wrong

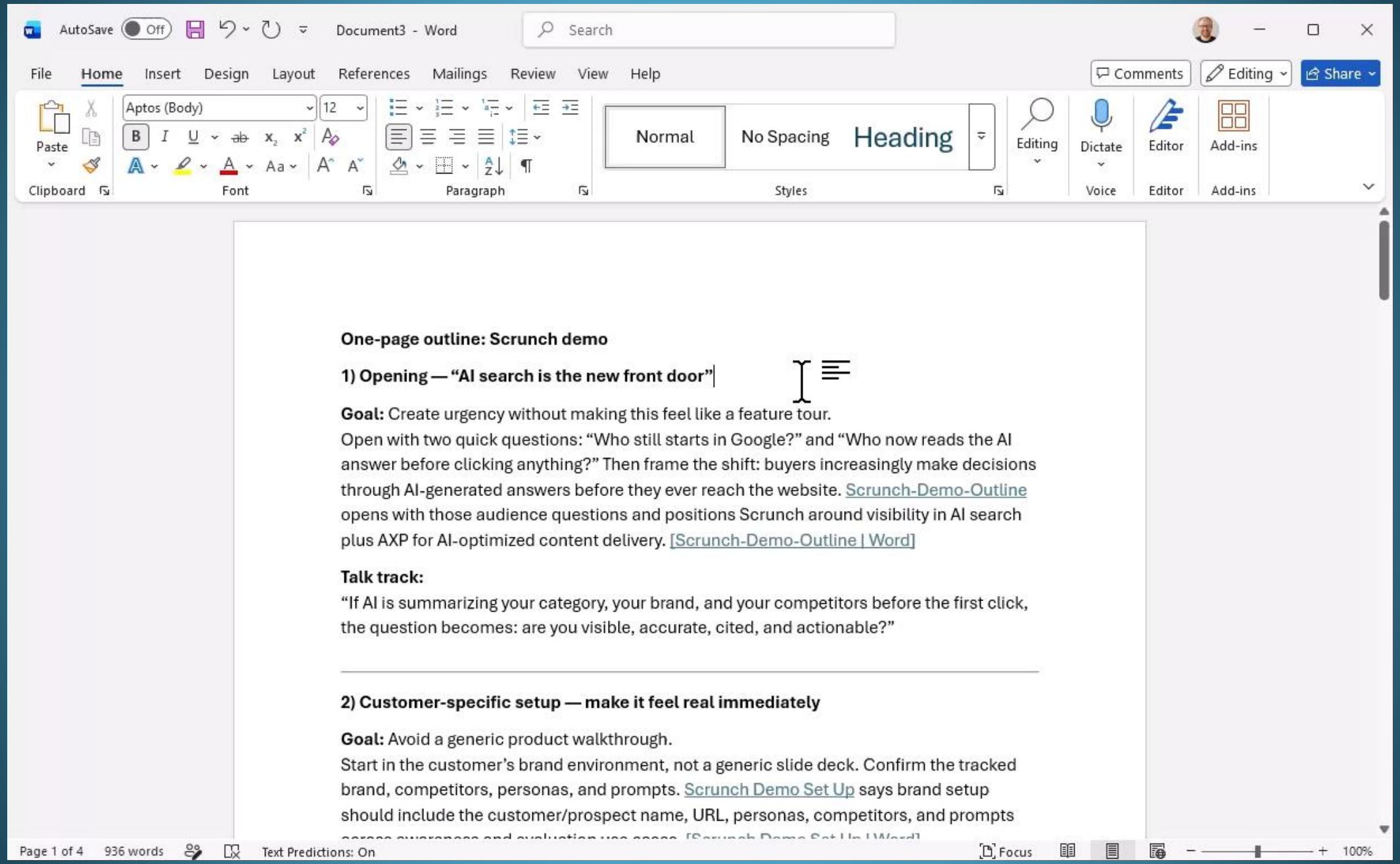
☑ **Yes**

? **Maybe Later**



Why So Many Clicks?




AutoSave Off
Document3 - Word
Search
File Home Insert Design Layout References Mailings Review View Help
Comments Editing Share
Paste
Aptos (Body)
12
Normal No Spacing Heading
Editing
Dictate
Editor
Add-ins
Clipboard Font Paragraph Styles Voice Editor Add-ins
One-page outline: Scrunch demo
1) Opening — "AI search is the new front door"
Goal: Create urgency without making this feel like a feature tour.
Open with two quick questions: "Who still starts in Google?" and "Who now reads the AI answer before clicking anything?" Then frame the shift: buyers increasingly make decisions through AI-generated answers before they ever reach the website. Scrunch-Demo-Outline opens with those audience questions and positions Scrunch around visibility in AI search plus AXP for AI-optimized content delivery. [Scrunch-Demo-Outline | Word]
Talk track:
"If AI is summarizing your category, your brand, and your competitors before the first click, the question becomes: are you visible, accurate, cited, and actionable?"
2) Customer-specific setup — make it feel real immediately
Goal: Avoid a generic product walkthrough.
Start in the customer's brand environment, not a generic slide deck. Confirm the tracked brand, competitors, personas, and prompts. Scrunch Demo Set Up says brand setup should include the customer/prospect name, URL, personas, competitors, and prompts
Page 1 of 4 936 words Text Predictions: On Focus 100%

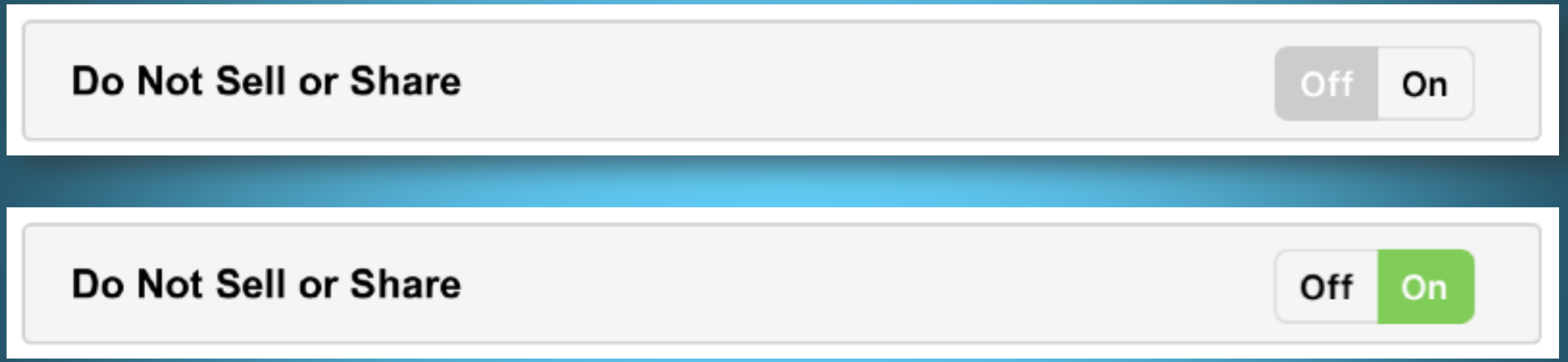
Flat Design



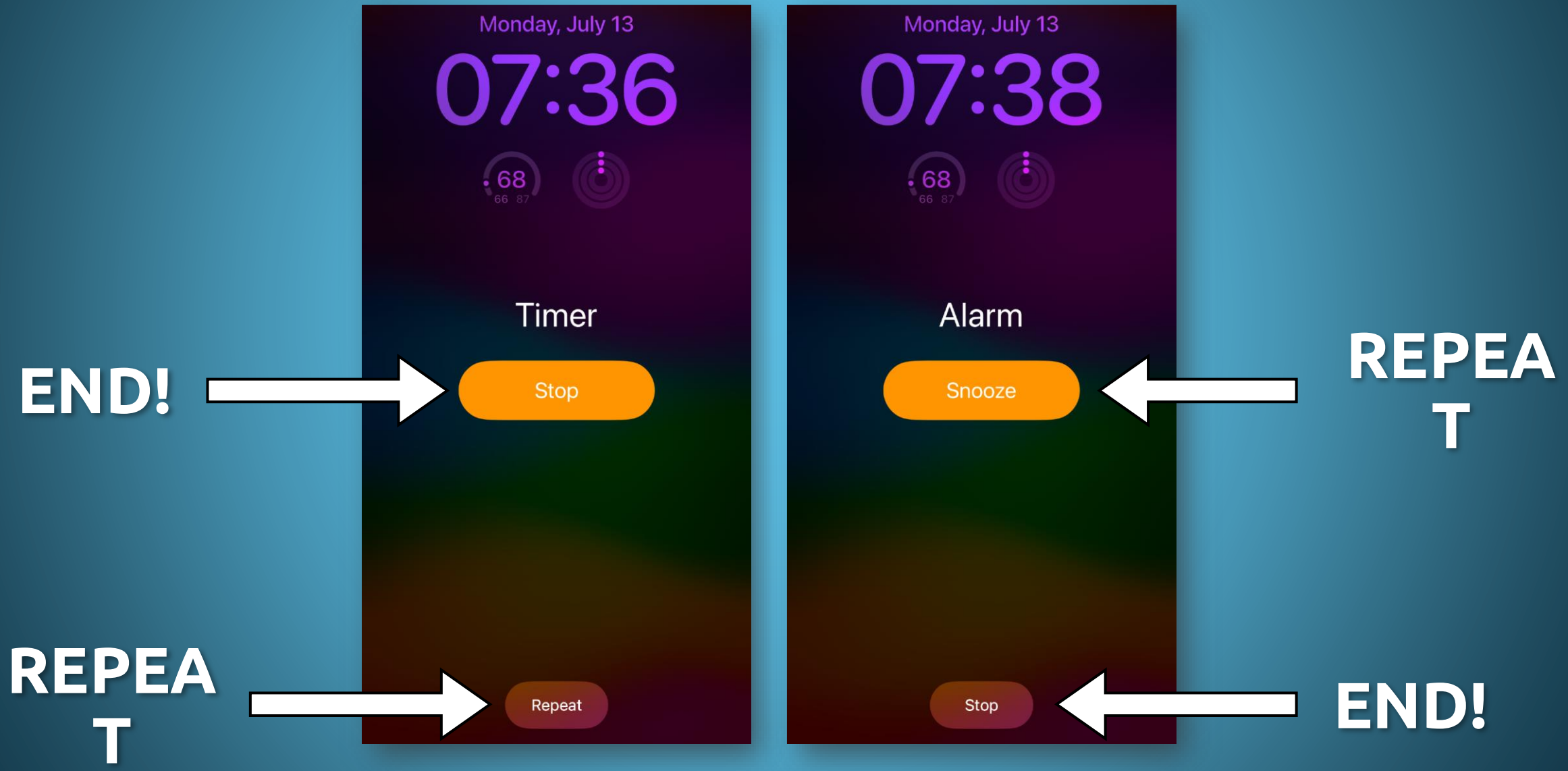
Submit

Reset

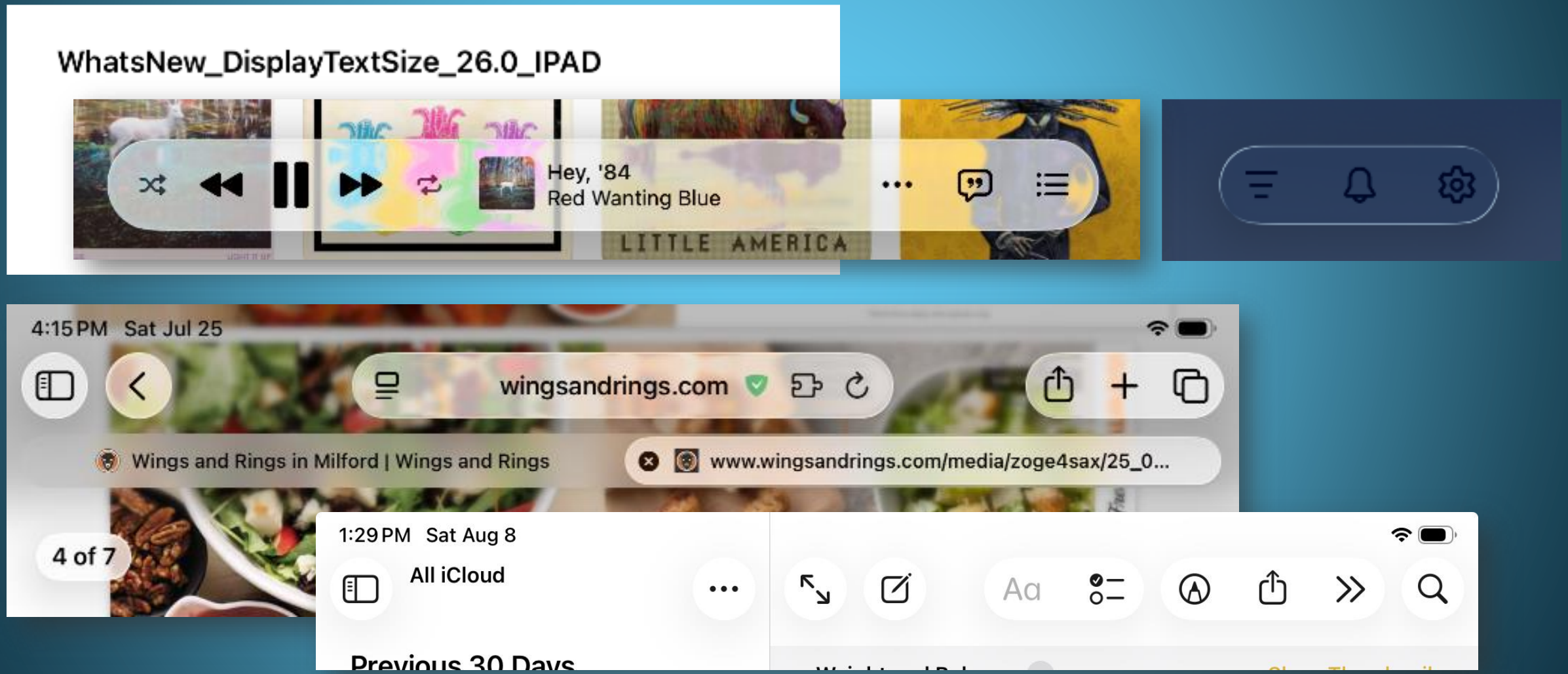
On or Off?



Same UI, Different Outcomes



Liquid Glass Gets Its Own Slide

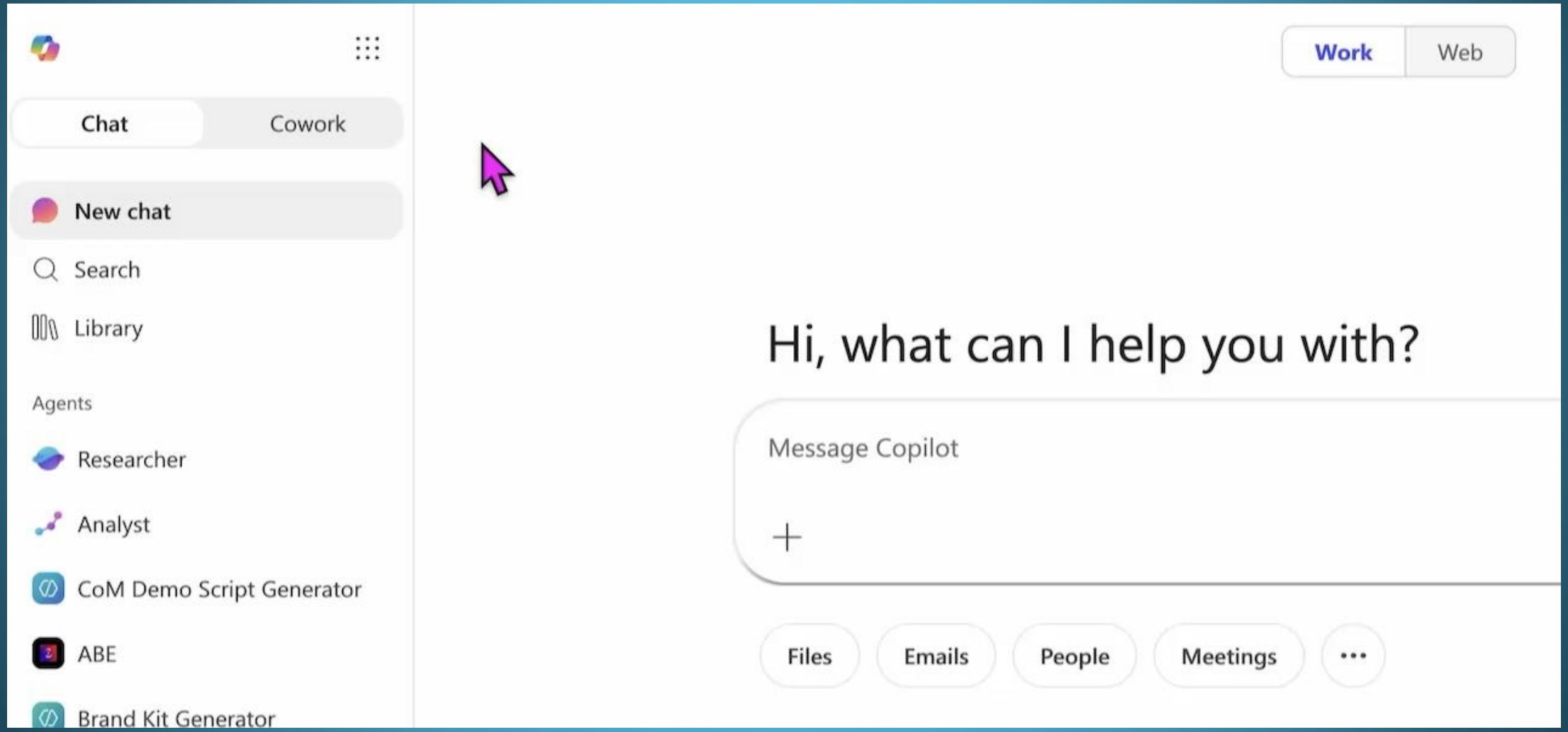


Regressions





Layout Shifting



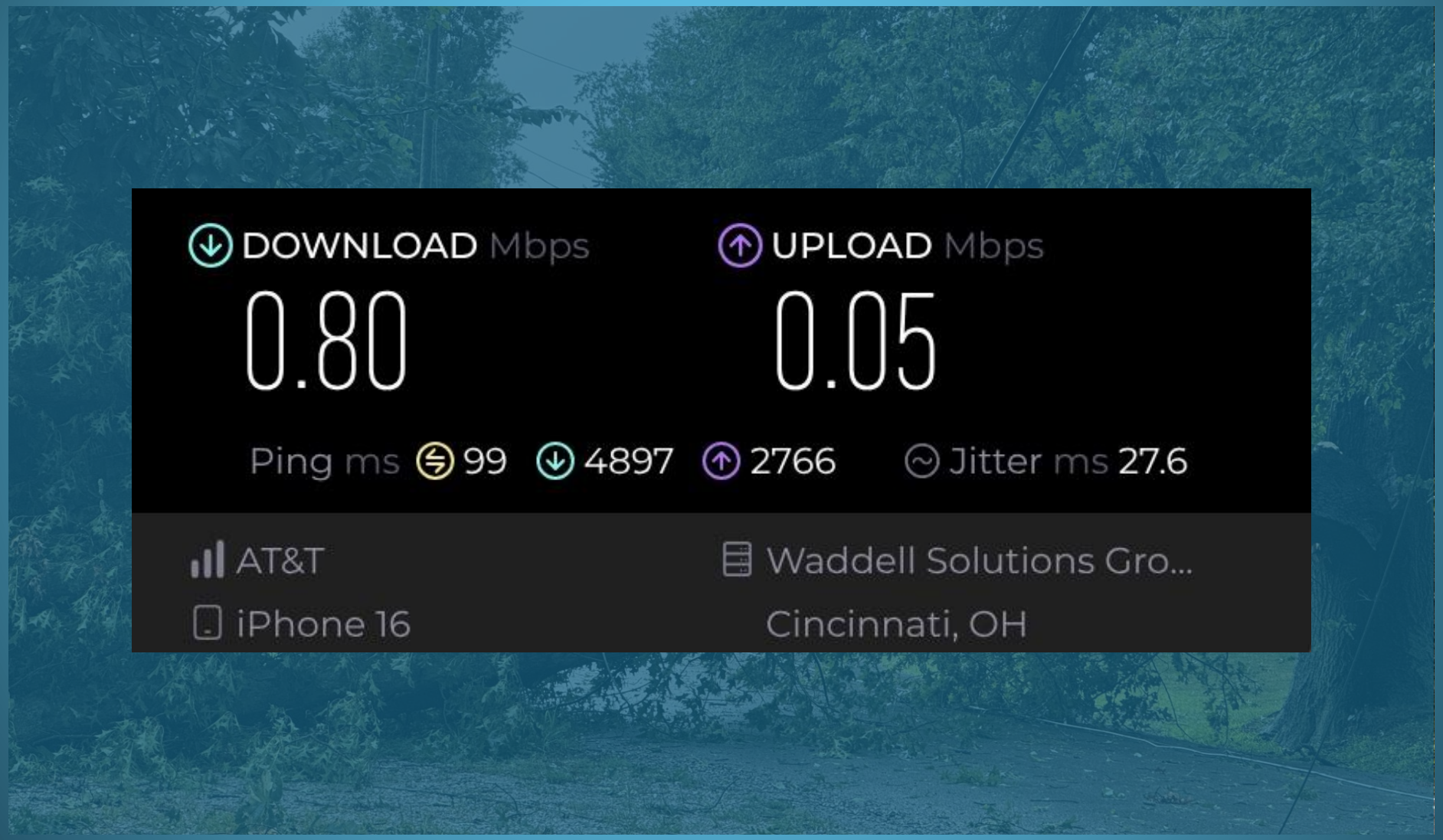
Network Lag as a Feature...?



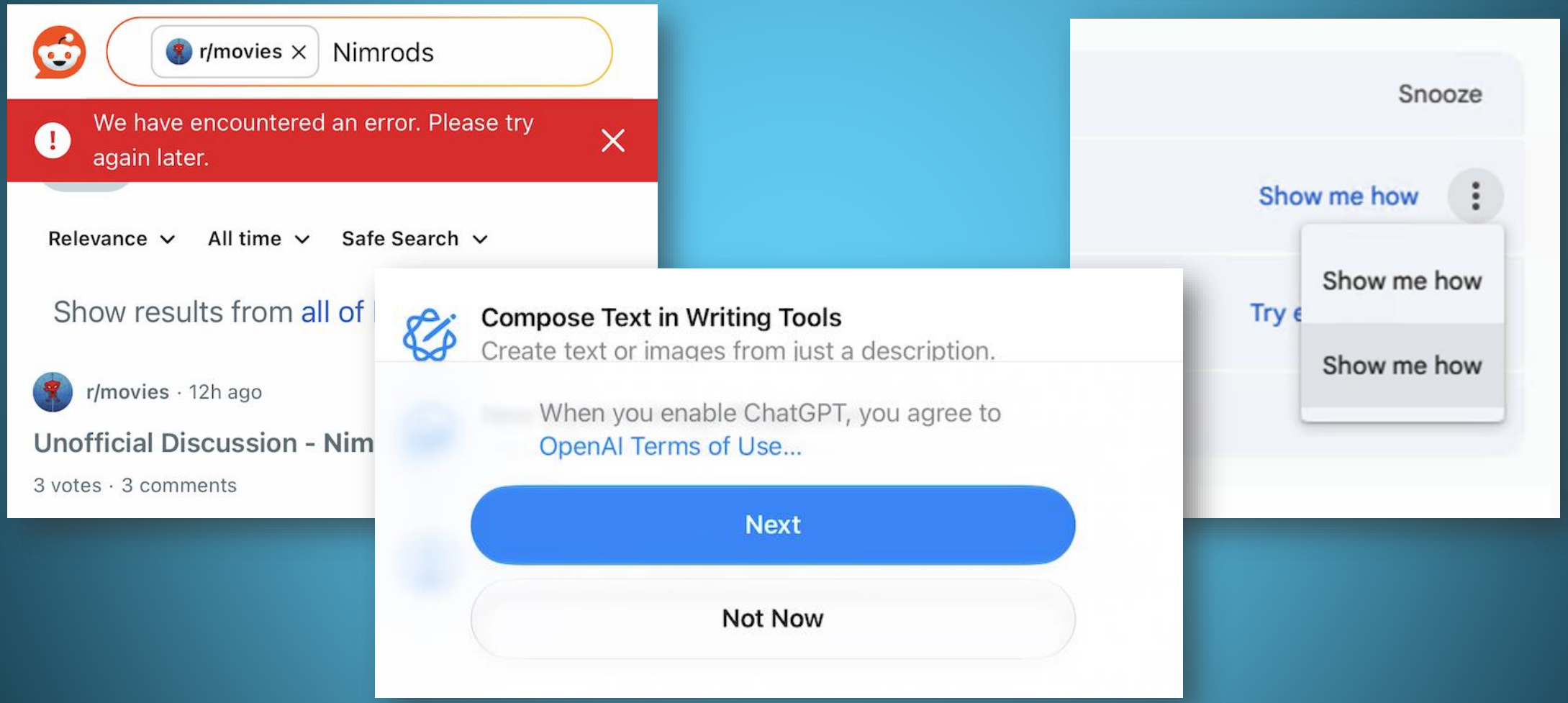




Dial-Up Still Exists (Thanks, AT&T)



But wait! What else?



So What Do We Do Now?

The Basics

You are not your user

We're technical.

We know the tool.

We're blind to user difficulties.

Boring is better

K.I.S.S.

Be predictable.

Don't overcomplicate.

Humans are the bottleneck

Humans have limited facilities.

Test, learn, and grow with users.

Discoverability	**Is the UI obvious? Are actions easy to find?**
Feedback	**What was the result of an action? Success or failure?**
Consistency	**Re-use of patterns; user doesn't have to re-learn**
Affordance	**Design of elements imply their use and purpose**
Cognitive Load	**The processing & memory burden placed on a user**

The UX Spectrum

Beautiful — **Functional**

PC — **Mac**

Dogs — **Cats**

Security — **Convenience**

Cincy Reds **???**



Remember Your Verbs


Click


Scroll


Tap


Hold

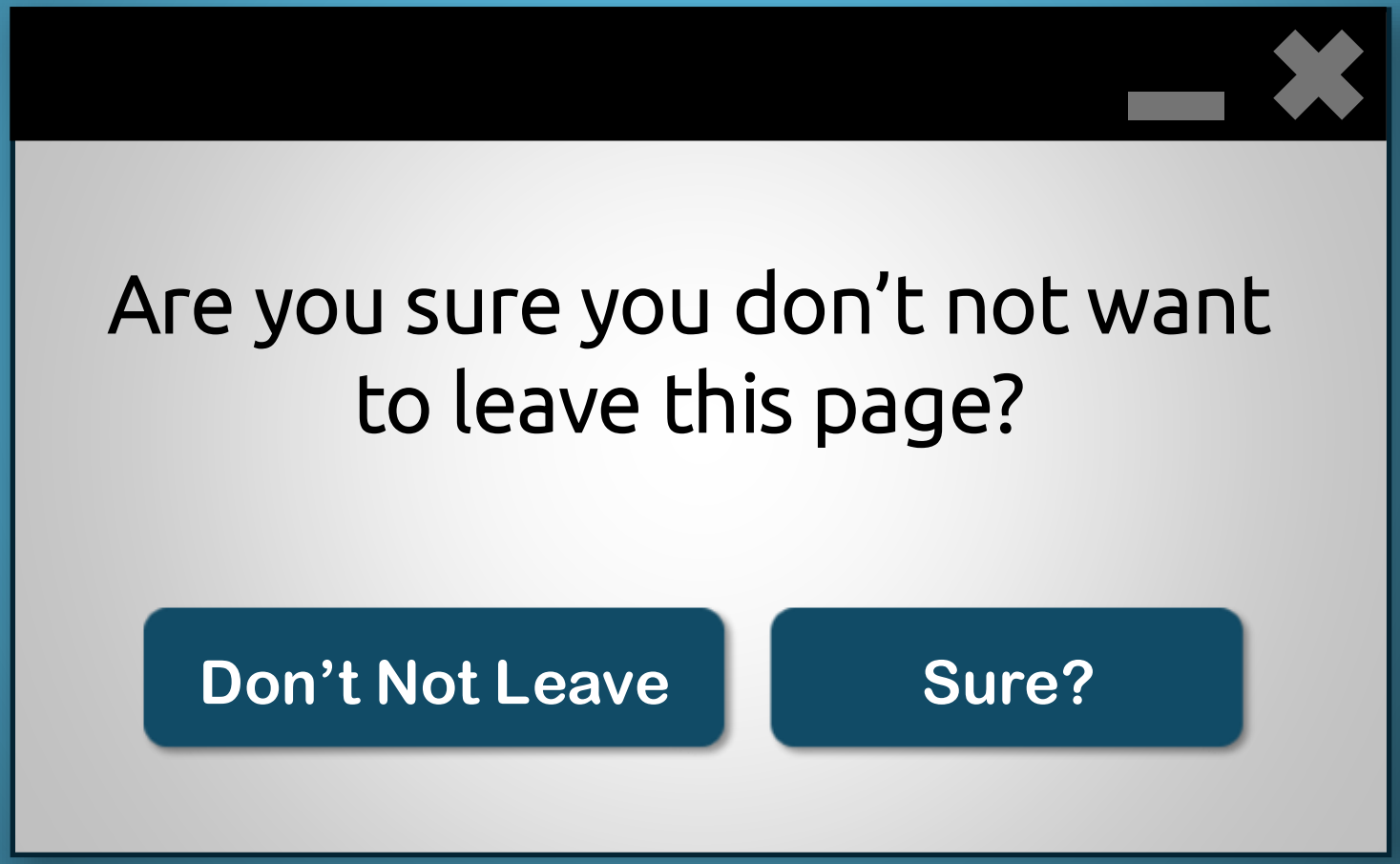

Drag


Pinch


Type


Swipe

With Great Power...



Accessibility Is Not Optional



Human-Computer Interaction

“Curb Cut Effect”

Handling Accessibility



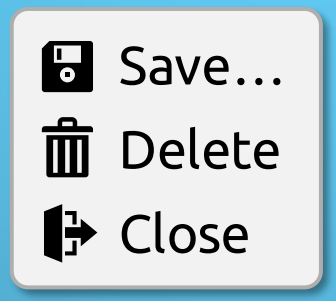
Native UI Design



Keyboard Navigation

This is fine.
Everything's fine.

Contrast



Icons + Text



Respect Zoom

1-800-HCI-LAWS

Jakob’s Law

Fitt's Law

Miller’s Law

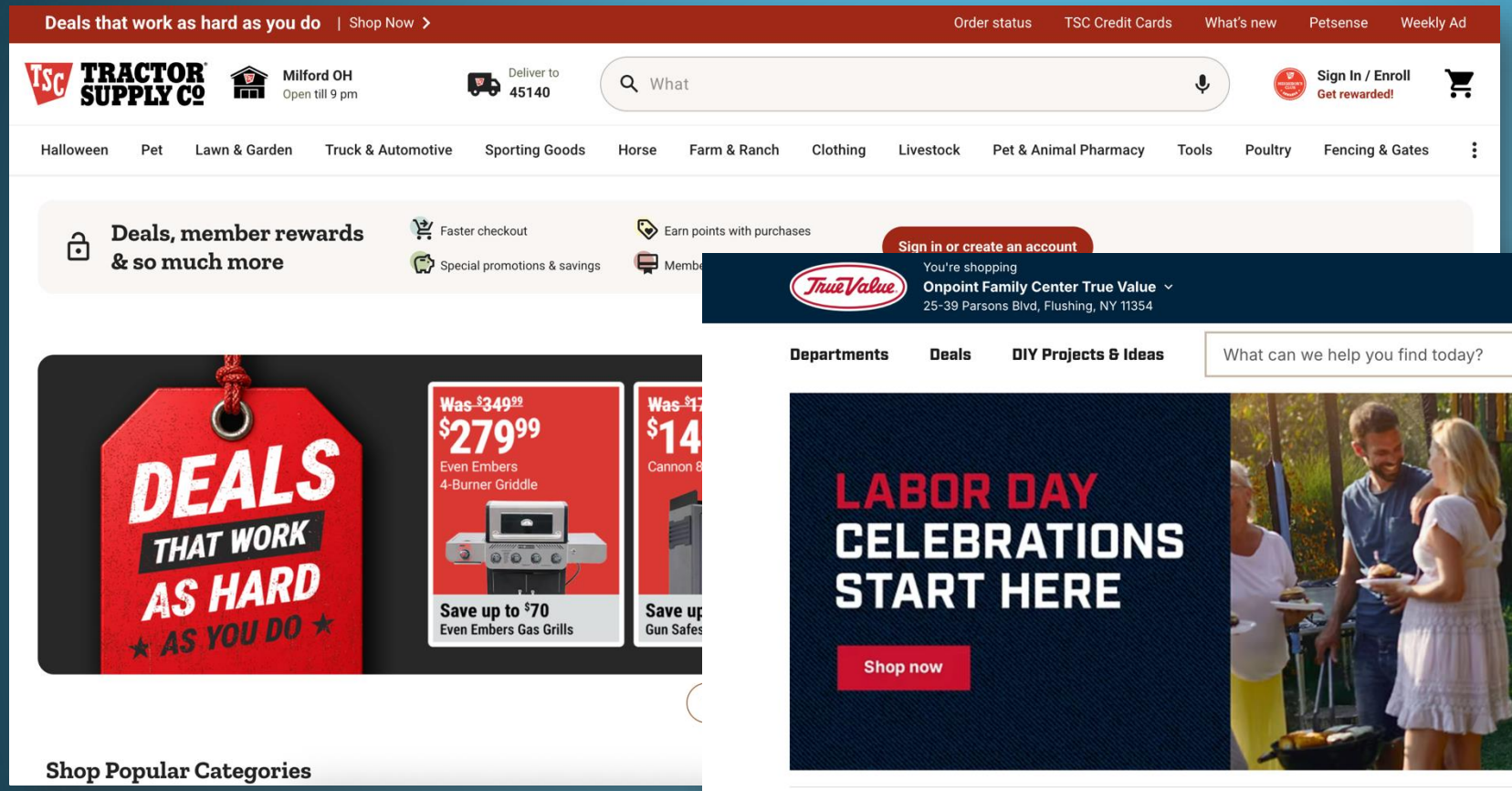
Hick-Hyman Law

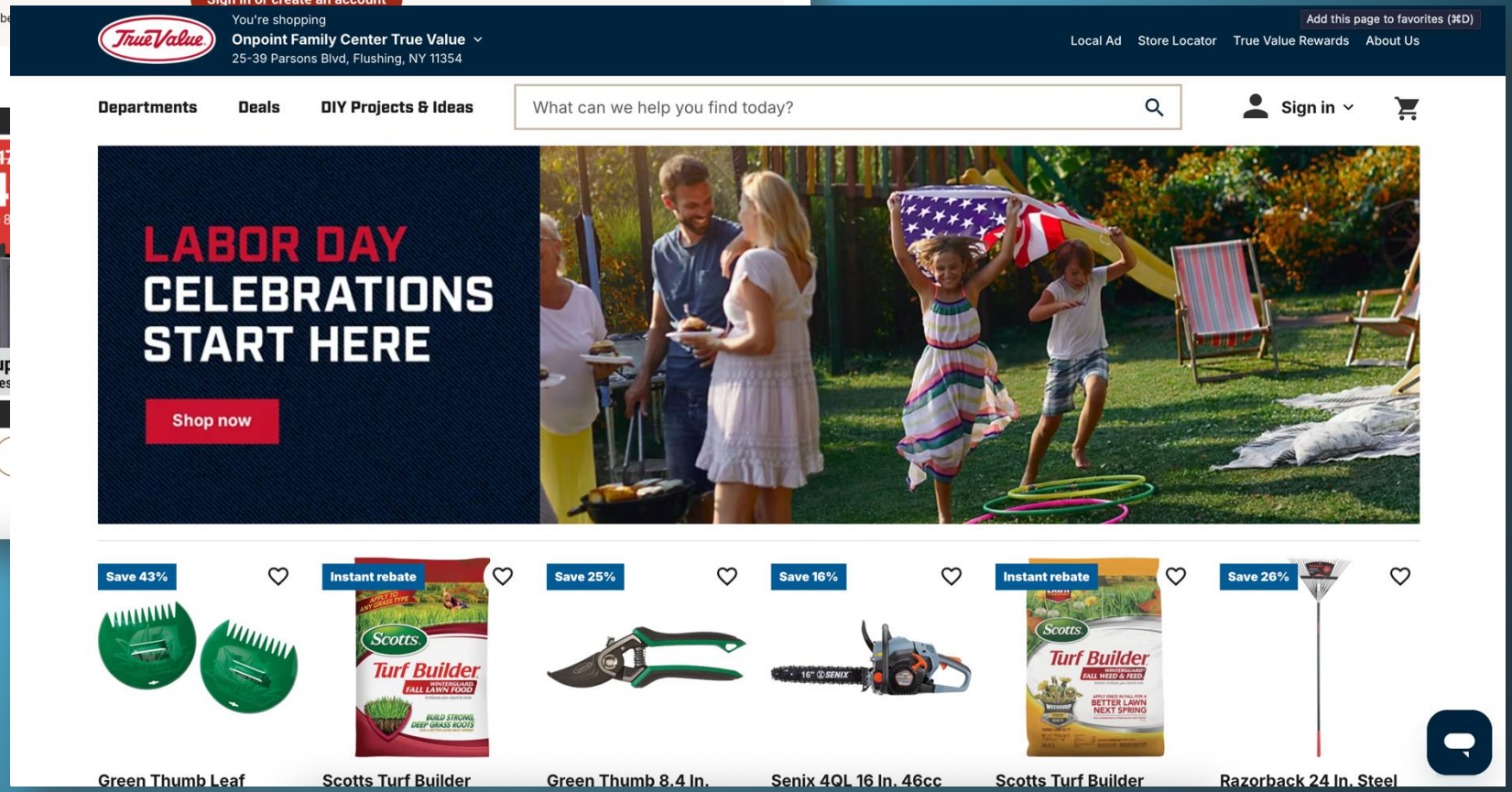
Power Law of Practice

Pareto Principle (not a law, I know)

Zipf Law

Jacob’s Law


Deals that work as hard as you do | Shop Now
Order status
TSC Credit Cards
What's new
Petsense
Weekly Ad
TRACTOR SUPPLY CO
Milford OH
Open till 9 pm
Deliver to
45140
What
Sign In / Enroll
Get rewarded!
Halloween
Pet
Lawn & Garden
Truck & Automotive
Sporting Goods
Horse
Farm & Ranch
Clothing
Livestock
Pet & Animal Pharmacy
Tools
Poultry
Fencing & Gates
Deals, member rewards & so much more
Faster checkout
Earn points with purchases
Special promotions & savings
Sign in or create an account
DEALS THAT WORK AS HARD AS YOU DO
Was $349.99
$279.99
Even Embers 4-Burner Griddle
Save up to $70
Even Embers Gas Grills
Shop Popular Categories

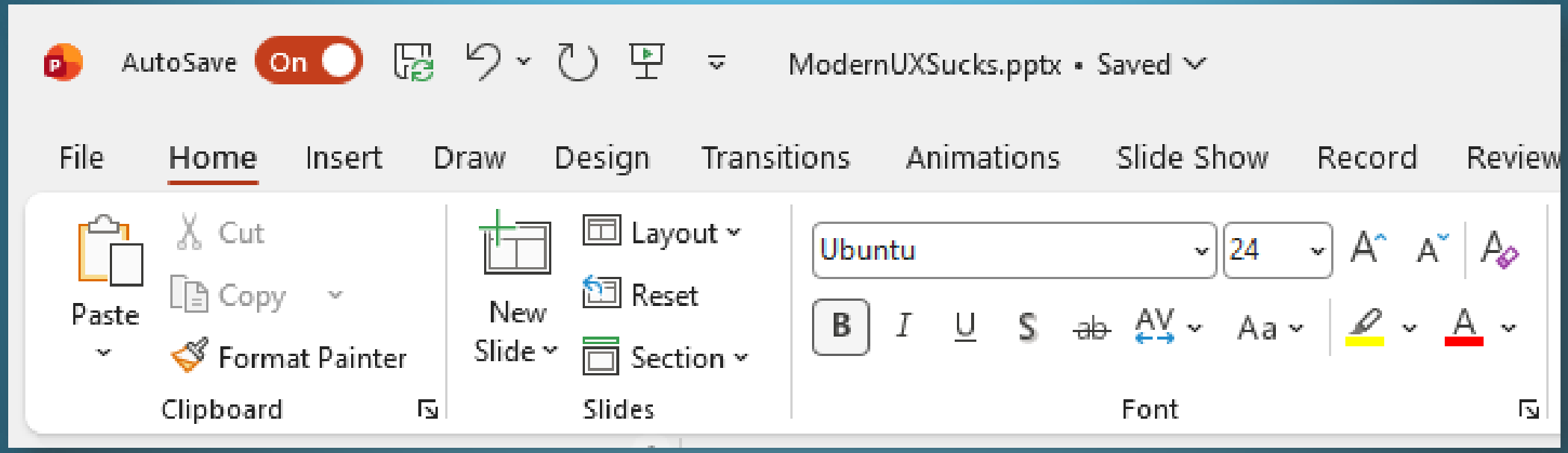

True Value
You're shopping
Onpoint Family Center True Value
25-39 Parsons Blvd, Flushing, NY 11354
Add this page to favorites (⌘D)
Local Ad
Store Locator
True Value Rewards
About Us
Departments
Deals
DIY Projects & Ideas
What can we help you find today?
Sign in
LABOR DAY CELEBRATIONS START HERE
Shop now
Save 43%
Instant rebate
Save 25%
Save 16%
Instant rebate
Save 26%
Scotts Turf Builder
Green Thumb Leaf
Scotts Turf Builder
Green Thumb 8.4 In.
Senix 40L 16 In. 46cc
Scotts Turf Builder
Razorback 24 In. Steel

Fitt's Law

- There's real math behind this



Pareto Principle & Zipf Law



Oh Yeah... That AI Thing

Evening thoughts

Read 2 pages, searched the web ›

Current dessert menu (Dolce, $8 each) from Sotto's official site:

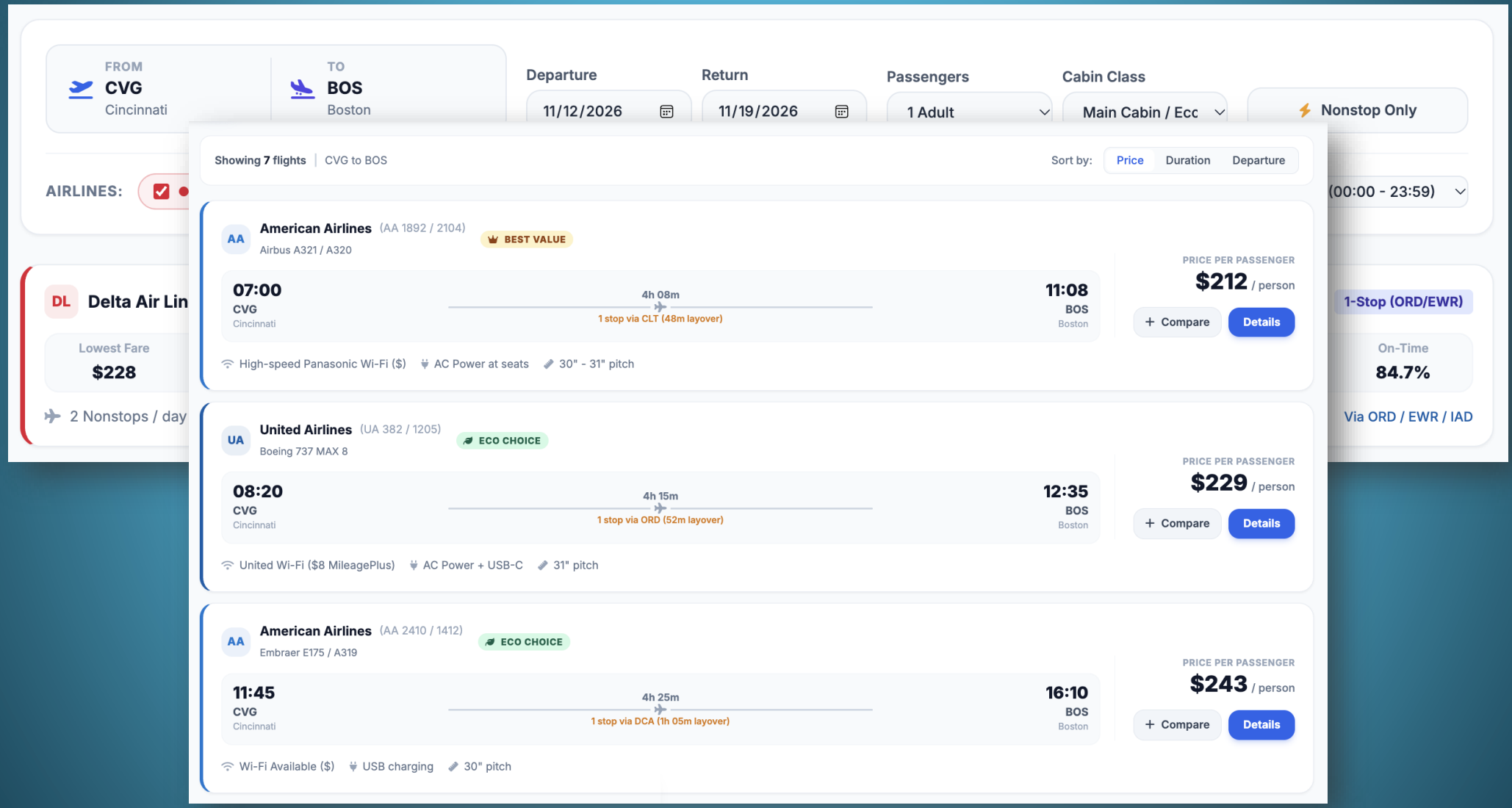
- Ricotta doughnuts with three sauces
- Chocolate budino with candied hazelnuts & gianduja mousse
- Buttermilk-vanilla panna cotta
- Selection of gelati or sorbetti

They also pair a $11 dessert wine flight (Vino) alongside — Recioto Della Valpolicella, Recioto di Soave, Fior d'Arancio, Vin Santo.

Menus rotate seasonally, so worth confirming with the restaurant for the current night's offerings.

Discoverability	**Describe an intention; bridging solves the gaps**
Feedback	**Natural language responses as part of UX**
Consistency	**Single UI/UX for interacting with many systems**
Affordance	**Multi-modal LLMs expose exact controls when needed**
Cognitive Load	**No mental models of UI; LLM responds directly**

You Get a UI, and You Get a UI!

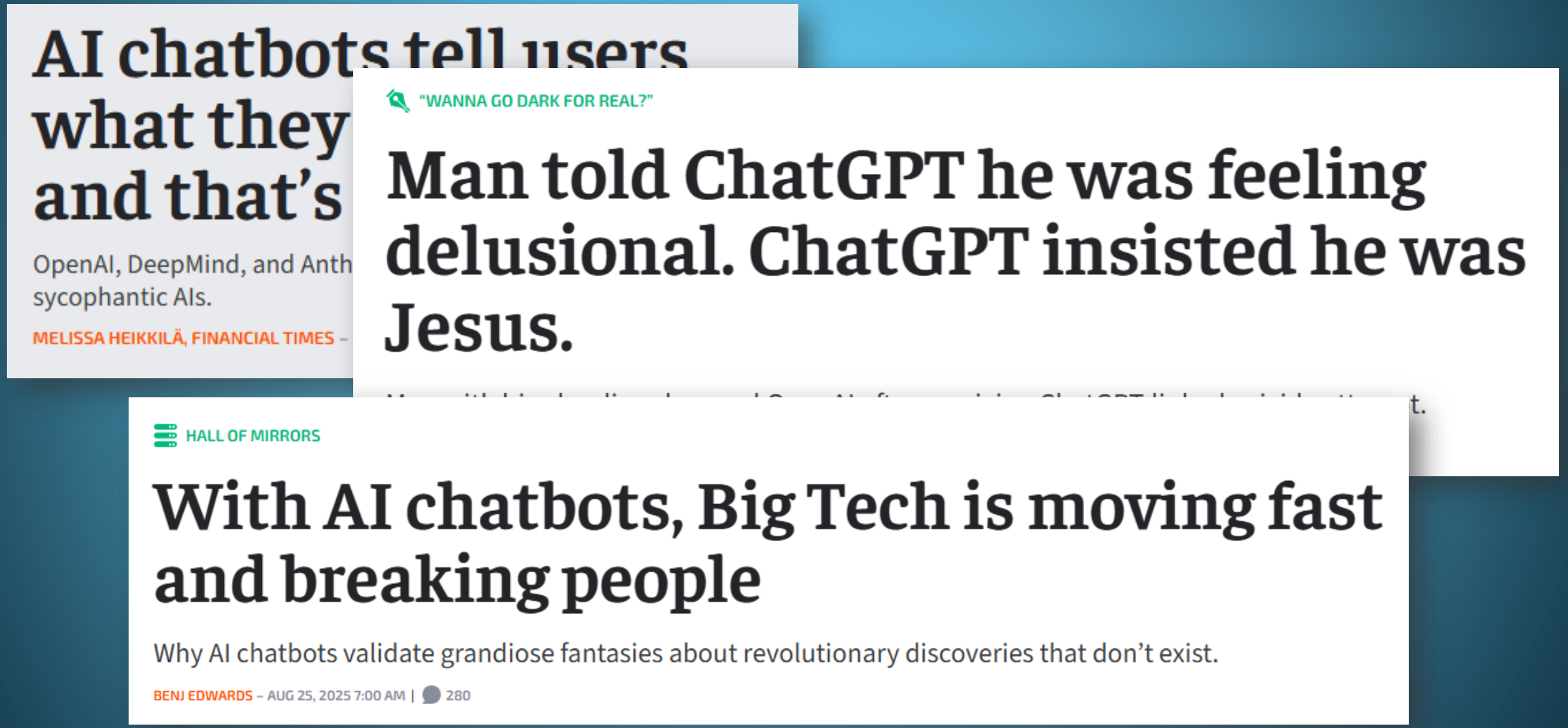


Say Hi to HAI

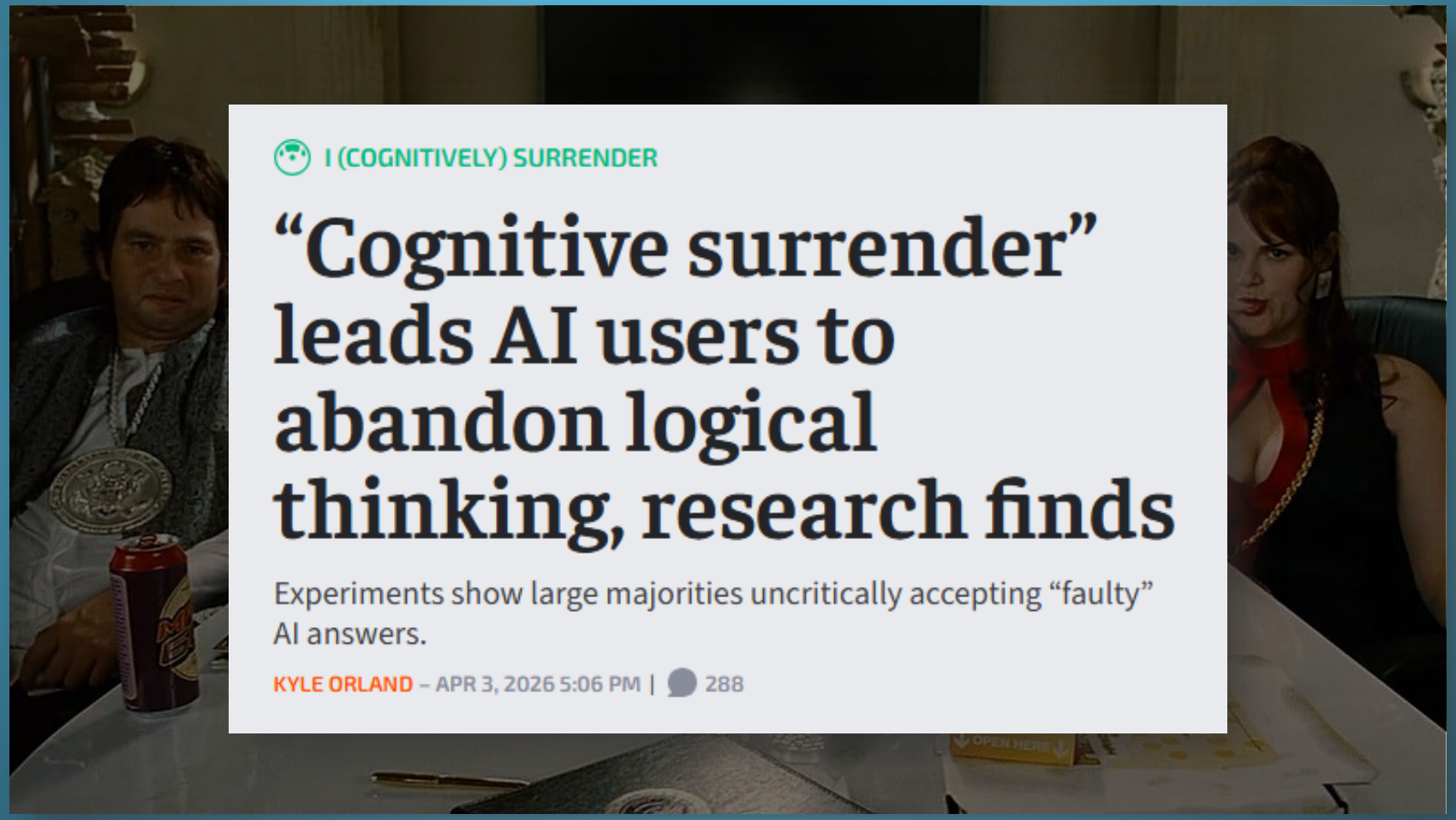
"**H**uman-Centered **A**rtificial **I**ntelligence"

- Values
- Needs
- Wellbeing
- Augmentation
- Transparency
- Safety
- Accessibility

The Dangerous Rabbit Hole



Welcome to Costco. I love you.



Attributions

- **Flaticons**: Magnific, IconsBox, Pixel perfect, alkhalifi design, Octopocto
- **Unsplash**: Ayako
- https://forum.winworldpc.com/discussion/15928/software-spotlight-30-years-of-windows-95
- https://www.typewritten.org/Media/

Thank You

 www.brandonbruno.com

 brandon.bruno@outlook.com

 /in/brandonbruno